

# Faculty of Computing

## SE1020 – Object Oriented Programming

### Year 1 Semester 2 (2025)

### Tutorial 03

---

#### Question 01 – Noun Verb Analysis

”ABC” Television Corporation is a leading television channel in Sri Lanka. The management has decided to automate the activities of their advertising department. Advertising department mainly focus on advertisement scheduling. You have been hired as a system analyst to identify the requirements and to make an initial sketch of the proposed system.

The staff of the advertising department consists with a manager, an assistant manager, a secretary and several designers. Before taking advertisements, ABC Television Corporation signs a contract with the client.

When a new advertisement is received from a client or a client company, the secretary will log in to the system and check the terms of the client contract by client number. Then the details of the advertisement ( This may include the length of the advertisement, time belt, payment, client name, client email etc) are entered in to the system. Since a new advertisement is available, secretary has to re-generate the advertisement schedule for the rest of the day. This schedule can be then printed and given to the telecasting unit.

If the client contract is changed due to any circumstance, the relevant advertisement schedule is updating accordingly. this is again done by the secretary.

An assistant manager can log in to the system and can view the advertisements added on a given date. For any reason, if the initially assigned time belt for a particular advertisement should be changed, that can only be done by an assistant manager or a manager.

At the end of each week, the assistant manager can generate the weekly scheduling status report. It visualize the status of the advertisement in the previous week. Frequently reports can be generate per each advertisement, to view the telecasting frequency of a particular advertisement. The client report can be generate at any time to get the details of clients; profit gained by each clients and the client payments. All the reports can be generated either by assistant manager or manager.

Clients can pay for advertisements directly through credit cards. The system is linked with existing accounting module to handle the payment details. An email will be send to the client for confirmation of payment. This will again handle by the accounting module.

**Do the Noun Verb analysis to identify the classes in the description.**

- a) Identify all the Nouns which can be potential classes
- b) Identify the set of potential classes, show clearly how the nouns were rejected using the five rules.
- c) Draw the CRC cards for any 5 classes you have identified in part (b)

### **Question 02 – Inheritance**

A Transport Company to develop a **Vehicle Management System**. The system manages different types of vehicles, such as cars and trucks, and calculates their rental costs.

The company applies different rental rates based on vehicle type as below:

| Vehicle Type | Rate per Day |
|--------------|--------------|
| Car          | \$50.00      |
| Truck        | \$80.00      |

- a) Create a class called **Vehicle** with the vehicleID (String), vehicleName (String), and rentalDays (int) as attributes.
- b) Implement a default constructor and a parameterized constructor.
- c) Implement a method called *displayDetails()* to show the vehicle information.
- d) Create a subclass called **Car** that inherits from Vehicle.
  - i. Implement a method *calculateRentalCost()* to calculate the rental cost using \$50.00 per day.
  - ii. **Override** the *displayDetails()* method to show the rental cost along with car details.
- e) Create another subclass called **Truck** that also inherits from Vehicle.
  - i. Implement a method *calculateRentalCost()* to calculate the rental cost using \$80.00 per day.
  - ii. Override the *displayDetails()* method to show the rental cost along with vehicle details.
- f) In the main() method, create objects of both Car and Truck, , and display their details using the overridden methods.

### Question 03 – Inheritance and Polymorphism

Implement an **Online Course Enrollment** System that manages different types of courses and calculates the total course fees for students. The platform offers both Regular Courses and Premium Courses. Premium courses include additional certification services, while regular courses do not.

- a) Create a class called **Course** with `courseID` (String), `courseName` (String), and `durationWeeks` (int) as attributes.
- b) Create a default constructor and a parameterized constructor to assign all details of a Course.
- c) Implement a method called *displayCourseDetails()* to print course information.
- d) Implement method overloading as below.
  - i. Implement *calculateTotalFee()* with no parameters to return 10000.00 as the basic course fee.
  - ii. Overload *calculateTotalFee* method with one double parameter as *discount-Percentage* to calculate the total fee using the discount value.  
*[Hint: call the calculateTotalFee() method get the basic course fee]*
- e) Create a subclass called **RegularCourse** that inherits from Course class and do followings.
  - i. Add an additional attribute `hasLiveSessions` (boolean) to indicate whether live sessions are included.
  - ii. Implement a default constructor and a parameterized constructor to assign all details of a Regular Course (including `courseID`, `courseName` and `durationWeeks`).
  - iii. Override the *calculateTotalFee()* method to compute the total fee as;  
$$\text{Total Fee} = 100 \times \text{durationWeeks}$$
  - iv. Override the *displayCourseDetails()* method to display all the details including the total fee and live session status.
- f) Create another subclass called **PremiumCourse** that inherits from Course and do followings.
  - i. Add an additional attribute called `includeCertification` (boolean).
  - ii. Implement a default constructor and a parameterized constructor to assign all details of a Premium Course (including `courseID`, `courseName` and `durationWeeks`).
  - iii. Override the *calculateTotalFee()* method to compute the total fee as;

$$\text{Total Fee} = (150 \times \text{durationWeeks}) + (\text{Certification Fee of \$50 if includeCertification is true})$$

- iv. Override the *displayCourseDetails()* method to include the total fee and certification status.
- g) Implement the main method as follows;
  - i. Create one object from **RegularCourse** using the default constructor.
  - ii. Create one **PremiumCourse** object using parameterized constructors.
  - iii. Call both overloaded versions of calculateTotalFee() for the objects.
  - iv. Display all course details using the overridden displayCourseDetails() method (refer to the output given below).

### Sample Output:

```
=== Regular Course (Default Constructor) ===
Course ID: N/A
Course Name: N/A
Duration: 0 weeks
Live Sessions Included: false
Total Fee: $0.0
Basic Fee: $0.0
Discounted Fee (10%): $0.0

=== Premium Course (Parameterized Constructor) ===
Course ID: PC202
Course Name: Machine Learning
Duration: 6 weeks
Certification Included: true
Total Fee: $950.0
Basic Fee: $950.0
Discounted Fee (20%): $760.0
```

### Question 04 – Inheritance and Polymorphism